

When Vocabulary-Row Proxies Fail: A Negative Result for Calibration-Free Vocabulary Quantization

Anonymous ACL submission

Abstract

Compressing the embedding and output projection matrices of large-vocabulary language models is attractive, but it is not an untouched deployment gap: practical quantizers already assign these tensors low-bit formats. We therefore test a narrower hypothesis: whether a completely calibration-free selector, using only tokenizer statistics, public token frequencies, row norms, and public entity-title membership, can choose high-precision vocabulary rows that preserve rare factual behavior better than random row bypass under the same vocabulary-byte budget. On a completed Qwen3-8B matrix spanning WikiText-2, TriviaQA rare-entity QA, Natural Questions rare-entity QA, MMLU, ARC-Challenge, and HellaSwag, the answer is negative. The best selector diagnostics are numerically close to random bypass controls: TriviaQA exact match is 41.00% versus a 40.73% random mean, Natural Questions is 13.00% versus 13.20%, and the lm-eval macro average is 72.22% versus 71.90%. Paired sign tests yield 0 significant random-control wins at $p < 0.05$, while WikiText-2 perplexity has only aggregate condition rows and is descriptive. The result does not make vocabulary quantization unimportant; it shows that public frequency/entity/norm proxies are too weak to support a positive long-tail factuality claim without calibration or stronger evidence.

1 Introduction

Vocabulary matrices are a visible memory component in modern decoder-only language models. The input embedding and output projection grow with vocabulary size, and models such as Qwen3-8B are deployed through frameworks that already expose memory/latency tradeoffs for both dense inference and quantized serving (Qwen Team, 2025; Wolf et al., 2020; GGML community, 2023). This makes the naive premise that embedding and output heads are ignored by deployment quantizers

untenable. The useful question is instead narrower: when those matrices are compressed under a fixed byte budget, can the rows that deserve extra precision be identified without forward calibration?

This paper studies that question as a failed-proxy experiment rather than as a success story. The intended mechanism is deliberately austere. It sees the tokenizer, public frequency counts, public entity-title token membership, and static row norms from the vocabulary matrices. It does not inspect validation answers, model predictions, loss traces, activation statistics, or calibration forward passes. That design makes the selector operationally cheap and leakage-resistant, but it also forces the core assumption into the open: high-frequency or high-norm entity-like rows must coincide with rows that matter for long-tail factual recall.

The distinction matters because vocabulary-row selection is unusually easy to overinterpret. A row can look rare, entity-like, or geometrically salient while still being irrelevant to the exact answer tokenization used by a QA benchmark. Conversely, an answer can depend on a common subword, a punctuation-sensitive continuation, or a decoder state that is not localized to one protected vocabulary row. A calibration-free policy is attractive precisely because it avoids forward passes, yet that same restriction removes the strongest evidence that a row actually changes model behavior. The experiment therefore treats public metadata as a hypothesis about row importance, not as a substitute for behavioral validation.

The completed evidence rejects the broad version of that assumption. Across 18,934 scored rows from six raw benchmark families plus a derived lm-eval macro view, the strongest selector diagnostics do not consistently separate from random bypass controls. The most favorable rare-entity row reaches 41.00% exact match on TriviaQA versus a 40.73% random-control mean, but the same entity-title plus norm selector reaches 13.00% on

Natural Questions versus a 13.20% random mean; the derived lm-eval macro row is 72.22% versus 71.90%. Paired exact sign tests over frozen task ids find 0/18 random-control comparisons with $p < 0.05$, and WikiText-2 perplexity remains descriptive because the evaluation stores one aggregate row per condition.

The contribution is therefore a negative one: for Qwen3-8B under this byte-matched vocabulary-matrix setting, public metadata plus row geometry is not enough to justify a positive long-tail factuality claim. The paper contributes (i) a leakage-controlled protocol for testing calibration-free row selection, (ii) a cross-benchmark matrix that keeps random bypass and same-backend FP16 context visible, and (iii) a claim-audit framing that prevents isolated diagnostic gains from becoming unsupported method claims.

2 Related Work

Post-training quantization. Modern LLM quantization already includes activation-aware, outlier-aware, second-order, smoothing, and mixed-precision designs. LLM.int8(), ZeroQuant, GPTQ, and SmoothQuant show that matrix outliers, layer-wise reconstruction, and activation scaling matter for low-bit inference (Dettmers et al., 2022; Yao et al., 2022; Frantar et al., 2022; Xiao et al., 2022). AWQ, OmniQuant, SpQR, and SqueezeLLM further demonstrate that calibration signals, salient channels, sparse outliers, and dense/sparse decompositions are central to strong post-training quantization (Lin et al., 2023; Shao et al., 2023; Dettmers et al., 2023b; Kim et al., 2023). Codebook and lattice methods such as additive quantization, QuIP#, and TesseraQ complicate any weak baseline story for vocabulary matrices (Egiazarian et al., 2024; Tseng et al., 2024; Li and Panda, 2024). Our negative result is consistent with this literature: a calibration-free vocabulary selector has to beat strong low-bit practice, not an imaginary FP16-only status quo.

Calibration-free and recent quantization variants. Recent work continues to explore microscaling formats, activation sensitivity, calibration-free subspace quantization, quantization-aware adaptation, reasoning-token side effects, optimized GPTQ variants, fairness-aware quantization, and geometric interpretations of GPTQ (Sharify et al., 2024; Xu, 2026; Qiao et al., 2026; Zhang and Saab, 2026; Lian et al.,

2026; Kong et al., 2025; Proskurina et al., 2025; Chen et al., 2025). These systems motivate the operational appeal of avoiding calibration, but they also raise the bar for evidence: simplicity alone is not a paper-facing accuracy claim. Our protocol isolates one simple metadata-only selector and asks whether it survives random-bypass controls under a fixed vocabulary-byte budget.

Embedding, codebook, and rare-token structure. Embedding compression has a long history outside LLM vocabulary heads. Vector quantization and product quantization provide classic codebook-sharing tools (Linde et al., 1980; Jégou et al., 2011), while recommendation systems have used large embedding-table quantization and frequency-skewed access patterns as first-class concerns (Naumov et al., 2019; Guan et al., 2019). Q-BERT and QLoRA illustrate additional low-precision directions in transformer compression and adaptation (Shen et al., 2019; Dettmers et al., 2023a). The closest motivation for this project is the observation that quantized embeddings can flatten spectral structure in frequency-sensitive ways (Huang et al., 2026). Our experiment tests a stricter claim: whether frequency/entity/norm proxies identify vocabulary rows that preserve rare factual answers better than random bypass.

Benchmarks and harnesses. The evaluation combines language modeling, rare-entity QA, and zero-shot multiple-choice tasks. WikiText-2 comes from the WikiText language-modeling corpus (Merity et al., 2016); rare-entity QA slices are derived from TriviaQA and Natural Questions (Joshi et al., 2017; Kwiatkowski et al., 2019); the zero-shot families use MMLU, ARC-Challenge, and HellaSwag through the lm-evaluation-harness interface (Hendrycks et al., 2021; Clark et al., 2018; Zellers et al., 2019; EleutherAI, 2024). These sources are intentionally heterogeneous: a selector with only one favorable aggregate PPL row would not be enough to support the long-tail factuality thesis.

3 Method

3.1 Vocabulary-Row Proxy

The evaluated proxy assigns each vocabulary row a score from three public or static signals. The first signal buckets tokenizer tokens by public corpus frequency, which approximates Zipfian exposure without touching the evaluation set. The second ranks rows by the norm of the embedding/output

Proxy Failure in Calibration-Free Vocabulary Quantization

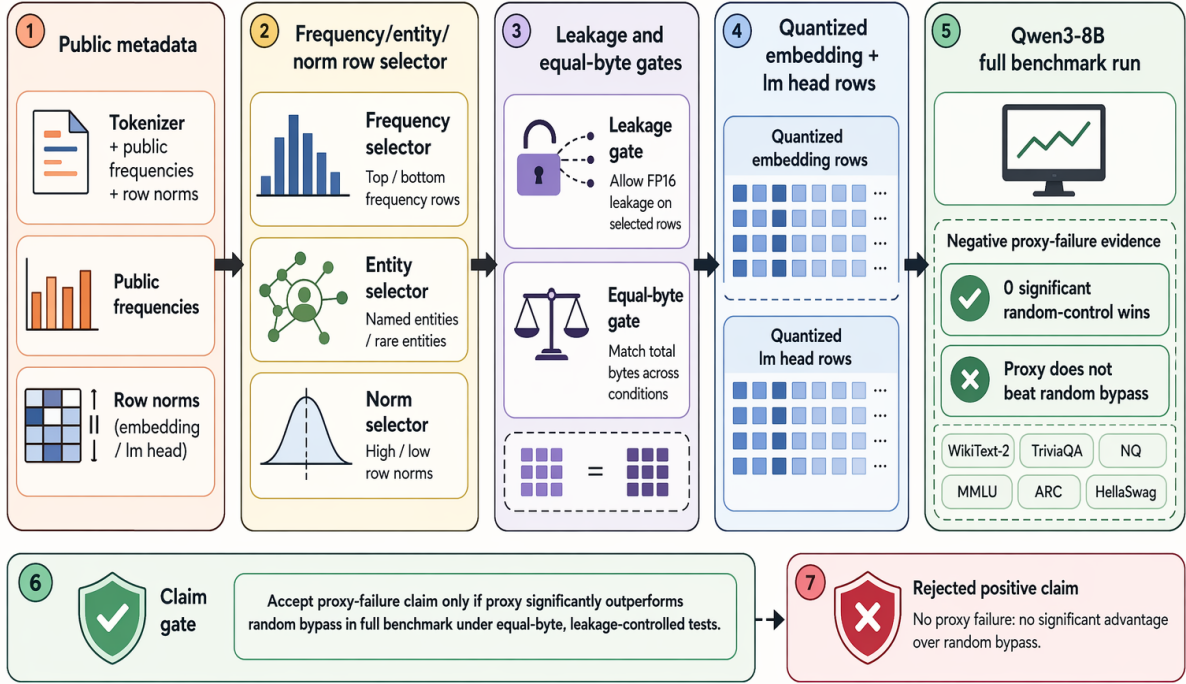


Figure 1: Calibration-free vocabulary-row selection is evaluated as a proxy test, not as a claimed win. Public metadata and row geometry feed the selector; equal-byte and leakage gates freeze admissible comparisons; the completed Qwen3-8B benchmark matrix yields 0 significant random-control wins.

vectors, using the intuition that unusually large rows may carry specialized lexical or factual information. The third marks tokens that occur in public entity titles, making a weak public-prior guess about rows likely to participate in rare entity answers. The selector combines these signals to reserve higher precision or residual-codebook capacity for a small row subset while quantizing the rest of the vocabulary matrix.

The important design constraint is what the selector cannot use. It cannot use benchmark labels, answer-token inventories, model generations, perplexity gradients, activation statistics, or calibration batches. Diagnostic answer-token oracle rows exist only to characterize ceiling behavior; they are not admissible methods. Random bypass controls are first-class baselines because the core question is whether the proxy selects better rows than a byte-matched random subset.

Operationally, the selector produces a frozen row set before any benchmark is scored and reuses that same row set across WikiText-2, QA, and lm-eval families. This prevents per-benchmark tuning after seeing outcomes and makes the random con-

trols meaningful: each random seed follows the same byte budget and bypass size, but replaces the metadata-driven ordering with an arbitrary row subset. The protocol therefore asks a sharp question. If the public proxy carries useful factuality information, it should separate from these random subsets on the frozen rare-entity and zero-shot tasks. If it does not separate, the evidence should be reported as proxy failure rather than as a small positive effect.

3.2 Byte and Backend Gates

Every paper-facing selector is interpreted under a fixed vocabulary-byte budget. The in-memory Hugging Face adapter rows use the same Qwen3-8B checkpoint and scoring codepath for QA and lm-eval. GGUF/K-quant rows are included only for WikiText-2 deployment context because they use a different backend and likelihood implementation; their absolute perplexities are not treated as cross-backend parity evidence. This distinction is binding for the claims: q4 rows can show that deployment formats already quantize vocabulary tensors, but they cannot serve as a QA or lm-eval

cross-benchmark control in the current matrix.

The byte gate also keeps the negative result from becoming a straw-man comparison. The method is not compared against a body-only compression setting with vocabulary tensors left untouched, because that would contradict current deployment practice. Instead, the main behavioral comparison is same-backend and equal-role: protected vocabulary rows selected by the proxy versus protected vocabulary rows selected at random. GGUF rows are kept in the paper because they calibrate the deployment premise, but their role is explicitly narrower than the random-bypass control.

3.3 Claim Gate

The analysis is pre-specified around a conservative claim gate. The central claim is negative: no reliable benefit over random bypass is observed, so the paper must not state selector superiority over random or equal-byte controls. The main scope claim is also narrow: all evidence is Qwen3-8B only, so cross-backbone generalization is not supported. This gate is part of the method because it prevents post-hoc narrative drift from turning isolated local diagnostics into a positive method result.

The claim gate affects both table design and prose. Main tables always show random context beside selector scores; captions state numerical negative-result headlines rather than only describing columns; and unsupported extrapolations are moved into limitations. The same gate also handles adverse comparisons. For example, the TriviaQA selector row has an isolated local diagnostic against same-backend FP16, but the random-control and Natural Questions evidence do not support a method claim. That local diagnostic is therefore discussed only as an anomaly that motivates proxy analysis.

4 Experimental Setup

Model and systems. All executed rows use Qwen/Qwen3-8B. The main in-memory evaluations use a Hugging Face causal-LM adapter that applies vocabulary-row quantization at the embedding/output matrices while leaving the benchmark harness fixed. WikiText-2 also includes GGUF llama.cpp context rows for FP16 and Q4-K-M vocabulary settings; those rows are restricted to backend-scope analysis. No training or gradient adaptation is performed.

Benchmarks and metrics. The full matrix covers WikiText-2 perplexity over 240 validation documents, TriviaQA and Natural Questions rare-entity exact match over 500 examples per family, and zero-shot MMLU, ARC-Challenge, and HellaSwag accuracy over 240 examples per family. The lm-eval macro row averages the three zero-shot families. Per-condition binary proportions use Wilson intervals, while paired exact sign tests compare frozen task ids for QA and lm-eval families. WikiText-2 has one aggregate row per condition, so PPL significance is not claimed.

Baselines and controls. The table matrix includes same-backend FP16 reference rows, deterministic row-norm and frequency controls, the entity-title plus norm selector, the high-norm residual-codebook selector, three random bypass seeds, and WikiText-only GGUF context rows. The paper-facing baseline for the main negative claim is the random bypass family, because the scientific question is whether public metadata selects better protected rows than an equally sized arbitrary bypass. Equal-byte GGUF rows remain deployment context until comparable generation/evaluation rows exist for QA and lm-eval.

Freezing and stopping rules. All source-family task ids are frozen before condition scoring. The QA slices use public rare-entity examples and greedy generation with a fixed maximum answer budget. The zero-shot families are evaluated through the same harness interface and then summarized both per family and as a macro average. Random bypass controls use three seeds, which is enough to expose whether the proxy is clearly separated from arbitrary row protection but not enough to support a broad small-effect claim. The evaluation stops after the predeclared matrix is complete; no row selector is reranked after seeing task outcomes.

5 Results

Figure 1 summarizes the evaluated proxy and the claim gate. Table 2 gives the main cross-benchmark matrix. The headline is not a win: the strongest selector diagnostics are numerically near random bypass controls and do not form a consistent significant pattern.

The rare-entity rows illustrate why the positive thesis fails. Entity-title plus norm reaches 41.00% EM on TriviaQA, compared with a random-

Source family	Frozen count	Evaluated backend	Metric	Paper-facing use
WikiText-2	240 docs	Qwen3-8B adapter; context	HF GGUF PPL	Descriptive LM stress; no paired PPL test.
TriviaQA rare entity	500 ex.	Qwen3-8B adapter	HF EM	Rare-entity QA with frozen ids and random bypass.
Natural Questions rare entity	500 ex.	Qwen3-8B adapter	HF EM	Independent rare-entity QA transfer check.
MMLU	240 ex.	Qwen3-8B adapter through lm-eval harness	HF Acc.	Zero-shot knowledge control.
ARC-Challenge	240 ex.	Qwen3-8B adapter through lm-eval harness	HF Acc.	Multiple-choice reasoning control.
HellaSwag	240 ex.	Qwen3-8B adapter through lm-eval harness	HF Acc.	Commonsense control; macro component.

Table 1: Evaluation protocol: six public source families are frozen before scoring; QA and lm-eval rows enter paired tests, while WikiText-2/GGUF rows stay descriptive.

control mean of 40.73%, but the same selector reaches 13.00% EM on Natural Questions, below the 13.20% random-control mean. The derived lm-eval macro score is 72.22% against a 71.90% random mean, but this small numerical gap is not enough to establish a reliable selector benefit under paired tests. The result is better read as proxy failure than as a small positive effect.

WikiText-2 is similarly limited. The best HF selector diagnostic is frequency-controlled norm at 18.1605 PPL, and the random range is 18.1607–18.1725. Because each condition contributes one aggregate PPL row, this comparison is descriptive only. Same-backend FP16 remains lower at 17.8694 PPL, and GGUF q4 rows live in a different backend context. The manuscript therefore does not claim PPL significance or cross-backend superiority.

The main matrix also shows that the negative result is not driven by a single hostile benchmark. TriviaQA is the most favorable rare-entity slice for the entity-title plus norm selector, yet the absolute gap over the random mean is less than one percentage point and the paired random-control tests do not clear the predeclared threshold. Natural Questions reverses the sign. The multiple-choice families are similarly small: MMLU, ARC-Challenge, and HellaSwag all stay within a narrow band around random bypass, and the macro average inherits that uncertainty. A positive claim would require a consistent direction across these families or an independently justified slice where the proxy is expected to apply. The evaluated matrix provides

neither.

This is why the table caption emphasizes significant random-control wins rather than the largest raw score. Reporting only the best selector row would obscure the random-bypass denominator and invite a false story about targeted entity preservation. Keeping the random mean and range beside each selector row makes the diagnostic conclusion harder but more defensible: the proxy can move scores, but the movements are not distinguishable from arbitrary protected-row choices under the tested budget.

6 Significance and Analysis

The main matrix reports the paired exact sign-test ranges for the entity-title plus norm selector against the three random bypass seeds. Across TriviaQA, Natural Questions, MMLU, ARC-Challenge, HellaSwag, and the pooled lm-eval macro view, none of the 18 random-control comparisons reaches $p < 0.05$. The expanded paired-test table is kept in the reproducibility appendix so the body can focus on the cross-benchmark negative-result matrix.

The failure mode is informative. The selector can occasionally pick rows that change individual answers, and TriviaQA even contains an isolated same-backend FP16 diagnostic in favor of the selector. The predeclared analysis does not convert that local fact into a method claim because random controls and Natural Questions do not corroborate it. Public entity-title membership is an attractive proxy for rare factual tokens, but the row-level evi-

Benchmark	Count	Metric	Best selector	diag- nostic	Score	Random context	bypass	Interpretation
WikiText-2	240 docs	HF PPL	Frequency- controlled norm		18.1605	18.1649 (18.1607– 18.1725)		Descriptive only; PPL has one aggregate row.
TriviaQA rare entity	500	EM	Entity-title norm	plus	41.00%	40.73% (40.60– 40.80)	(40.60– 40.80)	Random-control p=0.625– 1; no reliable benefit.
Natural Ques- tions rare en- tity	500	EM	Entity-title norm	plus	13.00%	13.20% (13.20– 13.20)	(13.20– 13.20)	Random-control p=1; no reliable benefit.
MMLU	240	Accuracy	Entity-title norm	plus	65.83%	65.83% (65.83– 65.83)	(65.83– 65.83)	Random-control p=1; no reliable benefit.
ARC- Challenge	240	Accuracy	Entity-title norm	plus	90.42%	90.28% (90.00– 90.42)	(90.00– 90.42)	Random-control p=1; no reliable benefit.
HellaSwag	240	Accuracy	Entity-title norm	plus	60.42%	59.58% (59.17– 60.42)	(59.17– 60.42)	Random-control p=0.375– 1; no reliable benefit.
lm-eval macro	3 tasks x 240 ex.; macro- average	Accuracy	Entity-title norm	plus	72.22%	71.90% (71.81– 72.08)	(71.81– 72.08)	Random-control p=0.453– 1; no reliable benefit.

Table 2: Main full-run matrix: 0 significant random-control wins support a negative result, not a positive selector-benefit claim. Random context reports mean and range over three seeds; WikiText-2 PPL is descriptive because each condition has one aggregate row.

dence here suggests that it is too coarse: many rare answers are fragmented across tokenizer pieces, and high-norm or entity-like rows do not uniquely identify the rows that decide exact-match success.

The analysis also separates byte matching from backend calibration. The GGUF q4 rows matter because they refute the premise that vocabulary tensors are normally left in FP16. They do not, however, establish a cross-benchmark equal-byte control: the current GGUF evidence is WikiText-only and uses a different likelihood backend. The same-backend HF PPL reference is 17.8694, the best HF selector diagnostic is 18.1605, and the GGUF Q4-K-M context rows are 8.9666–9.0466 under their own backend. A positive paper would need comparable GGUF generation rows for QA and lm-eval or a carefully calibrated bridge. This paper instead preserves the narrower negative claim.

A second lesson is that answer-aware diagnostics and deployment-safe selectors should not be conflated. An oracle that protects rows known to occur in gold answers would be useful for diagnosing whether vocabulary rows can matter at all, but it would leak benchmark information and would not be a deployable calibration-free method. The public entity-title prior is the non-leaking substitute tested here, and it is much weaker. The gap between these two objects is the core proxy failure: public metadata is not specific enough to identify the behavioral rows that decide frozen answer cor-

rectness.

The third lesson concerns effect-size interpretation. With 500 QA examples per family and 240 examples per zero-shot family, a one- or two-example swing can visibly change a percentage but still be operationally unconvincing. The paired exact tests show many ties because most conditions answer the same frozen items correctly or incorrectly. When non-tie counts are small, isolated wins should be treated as diagnostics of sensitivity rather than as evidence of robust row selection.

The fourth lesson is about the unit of intervention. The selector acts on vocabulary rows, but the evaluated behavior is produced by full decoding trajectories. A protected output row can change the probability of emitting one token, yet exact match depends on prompt interpretation, intermediate hidden states, answer formatting, and the sequence of subsequent tokens. Similarly, protecting an embedding row can preserve a token representation without guaranteeing that the model retrieves or formats the target fact. This mismatch between row-level intervention and sequence-level metric is a plausible reason that public row metadata fails to produce a stable QA signal.

The fifth lesson is methodological. Negative results in quantization are easy to hide by reporting only a favorable diagnostic comparator, especially when multiple selectors, tasks, and backend contexts are available. The analysis is intentionally

conservative: it treats the random-bypass family as the primary behavioral denominator and treats same-backend FP16, deterministic norm controls, and GGUF context as separate pieces of evidence. That separation keeps the paper from collapsing different questions into one apparent result. The supported conclusion is not that every row selector fails, but that this calibration-free public-proxy selector has not earned the factuality-preservation claim.

Finally, the analysis suggests what evidence would be needed to reverse the conclusion. A stronger future result would need either a calibrated signal that predicts row importance before task scoring, a public prior that is demonstrably aligned with answer-token usage across multiple datasets, or a systems result showing that row heterogeneity provides memory or throughput benefits independent of accuracy gains. It would also need to run the same selector family on additional tokenizers and backbones, because a vocabulary-row proxy can be sensitive to segmentation conventions. Those requirements are outside the current evidence and are therefore not claimed here.

7 Failure Cases

The saved row-level outcomes support a qualitative taxonomy even without printing long examples in the main body. First, rare entity aliases are often split across multiple subword tokens, so a row selector can protect one visible title token while missing the token that controls an answer span. Second, public entity-title priors are broad: they include many rows that are entity-like but irrelevant to the frozen QA slice. Third, row norms are not a factuality oracle; a large vector norm can reflect formatting, tokenization frequency, or embedding geometry rather than answer-bearing knowledge. Fourth, backend context can be misleading if GGUF and HF rows are mixed as though their absolute PPL values were directly comparable.

These failure categories explain why the negative result is not merely a matter of small effect size. The mechanism lacks a signal that is specific to the benchmark behavior it is meant to protect. Without forward calibration, answer-aware leakage, or a stronger public prior, the proxy cannot reliably distinguish useful bypass rows from arbitrary ones.

The taxonomy also helps delimit what the paper does not rule out. It does not show that vocabu-

lary matrices should never be compressed, since deployed quantizers already do so and the memory incentive remains real. It does not show that row-level mixed precision is useless, since a calibrated selector or a learned reconstruction objective could still identify rows that matter. It shows that the particular metadata-only proxy tested here is too weak for the factuality claim it was meant to support. That distinction keeps the negative result useful: it narrows the design space toward behavioral calibration, better public priors, or kernels that can justify row heterogeneity through systems gains rather than unsupported accuracy claims.

There is also a reproducibility lesson in the failure cases. The rejected positive story would have been easy to phrase around the intuitive link between rare entities and high-norm rows, but the frozen controls show why intuition is insufficient. The row-selection interface is intentionally simple: row sets are fixed before scoring, task ids are frozen, and every main table value follows from the completed evaluation matrix rather than from selective prose reporting. This makes the negative result inspectable and gives future work a clear target for comparison.

The row-level failures are distributed rather than concentrated in a single corner of the matrix. Token fragmentation is most visible in rare-entity QA, where a protected title-like row can fail to cover the continuation pieces that make an answer exact. Broad public entity priors create the opposite problem: they admit many plausible rows that never become decisive for the frozen examples. High-norm rows add another ambiguity because geometric salience can reflect common formatting tokens, tokenizer conventions, or projection-scale effects rather than answer-bearing factual content. The backend caveat is orthogonal to these proxy failures, but it matters for interpretation because a low-bit deployment row cannot be treated as a same-backend behavioral control unless the generation and likelihood path are matched.

These observations make the negative result more actionable than a single aggregate loss. A future selector would need to demonstrate that its protected rows are behaviorally coupled to answer formation, not merely that they are rare, entity-like, or large in norm. The current proxy does not provide that coupling under the frozen tasks. This is the reason the paper keeps the claim at the level of proxy failure: the experiments expose where public metadata stops being a reliable substitute for

calibration, but they do not rule out calibrated row selection or systems-driven heterogeneous kernels.

8 Conclusion

This paper tests a calibration-free vocabulary-matrix selector under a completed Qwen3-8B benchmark matrix and finds no reliable benefit over random bypass controls. The result downgrades the original positive story: public frequency, entity-title membership, and row norms are useful diagnostic signals, but they are not sufficient evidence for preserving long-tail factuality under equal vocabulary-byte budgets. Future work should treat calibration-free vocabulary selection as a hypothesis requiring stronger proxy validation, not as a deployable accuracy claim.

Limitations

The evidence is limited to Qwen3-8B. It supports a single-backbone negative result, not a universal statement about Llama, Gemma, or other tokenizers. Vocabulary segmentation differs across backbones, so the same public entity prior could select different subword rows even before any model behavior is measured. WikiText-2 PPL is aggregate-only, so it cannot support paired significance. The GGUF/K-quant rows are valuable deployment context but are WikiText-only and backend-scoped; the reproducibility appendix records the PPL values as context rather than parity evidence. Finally, non-uniform per-row vocabulary quantization primarily saves memory in this implementation; it does not provide a custom GPU decode kernel or speedup claim.

The random-bypass family also has limited precision. Three seeds are enough to test whether the proxy has an obvious large advantage over arbitrary protected rows, but they are not a tight estimate of every small random-subset fluctuation. The conclusion is therefore deliberately conservative: the evaluated proxy does not earn a positive long-tail factuality claim under the tested budget and tasks. A stronger claim would require additional backbones, more random seeds or a closed-form sampling analysis, and same-backend low-bit controls that extend beyond WikiText-2 into the QA and zero-shot families.

Ethical Considerations

The benchmarks are public language-model evaluation sources, and the rare-entity slices are used

to audit factual behavior rather than to profile individuals. The main ethical risk is overclaiming factuality preservation from weak proxies. The manuscript avoids that risk by reporting the negative result directly, keeping random controls visible, and declining to generalize beyond the evaluated backbone and benchmark matrix.

The evaluation also avoids releasing new sensitive examples: it reports aggregate exact-match, accuracy, and perplexity outcomes over public benchmark items, with row-selection conditions frozen before scoring. The intended reproducibility path is to replay the same public task ids and condition definitions, not to expose private user data or deploy a factuality-preservation claim in downstream systems.

References

- Jiale Chen, Yalda Shabanzadeh, Elvir Crnčević, Torsten Hoefer, and Dan Alistarh. 2025. [The geometry of llm quantization: Gptq as babai’s nearest plane algorithm](#). *arXiv preprint arXiv:2507.18553v4*.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. [Think you have solved question answering? try ARC, the AI2 reasoning challenge](#). *arXiv preprint arXiv:1803.05457*.
- Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. 2022. [Llm.int8\(\): 8-bit matrix multiplication for transformers at scale](#). *arXiv preprint arXiv:2208.07339v2*.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023a. [QLoRA: Efficient finetuning of quantized LLMs](#). *arXiv preprint arXiv:2305.14314*.
- Tim Dettmers, Ruslan Svirschevski, Vage Egiazarian, Denis Kuznedelev, Elias Frantar, Saleh Ashkboos, Alexander Borzunov, Torsten Hoefer, and Dan Alistarh. 2023b. [Spqr: A sparse-quantized representation for near-lossless llm weight compression](#). *arXiv preprint arXiv:2306.03078v1*.
- Vage Egiazarian, Andrei Panferov, Denis Kuznedelev, Elias Frantar, Artem Babenko, and Dan Alistarh. 2024. [Extreme compression of large language models via additive quantization](#). *arXiv preprint arXiv:2401.06118v4*.
- EleutherAI. 2024. [Language model evaluation harness](#). Open-source benchmark harness for language-model evaluation.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. 2022. [Gptq: Accurate post-training quantization for generative pre-trained transformers](#). *arXiv preprint arXiv:2210.17323v2*.
- GGML community. 2023. [llama.cpp](#). Open-source LLM inference and quantization software.
- Hui Guan, Andrey Malevich, Jiyan Yang, Jongsoo Park, and Hector Yuen. 2019. [Post-training 4-bit quantization on embedding tables](#). *arXiv preprint arXiv:1911.02079v1*.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). *International Conference on Learning Representations*.
- Junlin Huang, Wenyi Fang, Zhenheng Tang, Yuxin Wang, Xueze Kang, Yang Zheng, Bo Li, and Xiaowen Chu. 2026. [On the spectral flattening of quantized embeddings](#). *arXiv preprint arXiv:2602.00969v1*.
- Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. 2017. [TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension](#). In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611. Association for Computational Linguistics.
- H Jégou, M Douze, and C Schmid. 2011. [Product quantization for nearest neighbor search](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Sehoon Kim, Coleman Hooper, Amir Gholami, Zhen Dong, Xiuyu Li, Sheng Shen, Michael W. Mahoney, and Kurt Keutzer. 2023. [Squeezellm: Dense-and-sparse quantization](#). *arXiv preprint arXiv:2306.07629v4*.
- Jie Kong, Junxiang Zhang, Jiheng Xu, Yalong Li, Shouhua Zhang, Jiehan Zhou, Yuhai Liu, Peng Liang, Quan Zhang, and Luohan Jiang. 2025. [Opt-gptq: An optimized gptq combining sparse attention and quantization techniques](#). *arXiv preprint arXiv:2505.02351v2*.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. [Natural questions: A benchmark for question answering research](#). *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Yuhang Li and Priyadarshini Panda. 2024. [Tesseract: Ultra low-bit llm post-training quantization with block reconstruction](#). *arXiv preprint arXiv:2410.19103v1*.
- Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. 2026. [Quantization inflates reasoning: Token inflation as a hidden cost of low-bit reasoning models](#). *arXiv preprint arXiv:2606.25519v2*.
- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. 2023. [Awq: Activation-aware weight quantization for llm compression and acceleration](#). *arXiv preprint arXiv:2306.00978v6*.
- Y. Linde, A. Buzo, and R. Gray. 1980. [An algorithm for vector quantizer design](#). *IEEE Transactions on Communications*.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2016. [Pointer sentinel mixture models](#). *arXiv preprint arXiv:1609.07843*.
- Maxim Naumov, Dheevatsa Mudigere, Hao-Jun Michael Shi, Jianyu Huang, Narayanan Sundaraman, Jongsoo Park, Xiaodong Wang, Udit Gupta, Carole-Jean Wu, Alisson G. Azzolini, Dmytro Dzhulgakov, and Andrey Mallevich. 2019. [Deep learning recommendation model for personalization](#)

and recommendation systems. *arXiv preprint arXiv:1906.00091v1*.

Irina Proskurina, Guillaume Metzler, and Julien Velcin. 2025. [Fair-gptq: Bias-aware quantization for large language models](#). *arXiv preprint arXiv:2509.15206v2*.

Ye Qiao, Yian Wang, Zhiheng Chen, Hyounjun Kwon, and Sitao Huang. 2026. [Fasq: Flexible accelerated subspace quantization for calibration-free llm compression](#). *arXiv preprint arXiv:2605.04084v1*.

Qwen Team. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.

Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. 2023. [Omniquant: Omnidirectionally calibrated quantization for large language models](#). *arXiv preprint arXiv:2308.13137v3*.

Sayeh Sharify, Utkarsh Saxena, Zifei Xu, Wanzin Yazar, Ilya Soloveychik, and Xin Wang. 2024. [Post training quantization of large language models with microscaling formats](#). *arXiv preprint arXiv:2405.07135v3*.

Sheng Shen, Zhen Dong, Jiayu Ye, Linjian Ma, Zhewei Yao, Amir Gholami, Michael W. Mahoney, and Kurt Keutzer. 2019. [Q-bert: Hessian based ultra low precision quantization of bert](#). *arXiv preprint arXiv:1909.05840v2*.

Albert Tseng, Jerry Chee, Qingyao Sun, Volodymyr Kuleshov, and Christopher De Sa. 2024. [Quip#: Even better llm quantization with hadamard incoherence and lattice codebooks](#). *arXiv preprint arXiv:2402.04396v2*.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2022. [Smoothquant: Accurate and efficient post-training quantization for large language models](#). *arXiv preprint arXiv:2211.10438v7*.

Bruce Changlong Xu. 2026. [Activation sensitivity as a unifying principle for post-training quantization](#). *arXiv preprint arXiv:2601.11663v1*.

Zhewei Yao, Reza Yazdani Aminabadi, Minjia Zhang, Xiaoxia Wu, Conglong Li, and Yuxiong He. 2022. [ZeroQuant: Efficient and affordable post-training quantization for large-scale transformers](#). *arXiv preprint arXiv:2206.01861*.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. [HellaSwag: Can a machine really finish your sentence?](#) In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4791–4800. Association for Computational Linguistics.

Shihao Zhang and Rayan Saab. 2026. [Gptq-intrinsic lora: A near-optimal algorithm for low-precision quantization with low-rank adaptation](#). *arXiv preprint arXiv:2606.01412v1*.

A Reproducibility Notes

The supplementary package records the run configuration, frozen task-id coverage, condition-equivalence checks, result tables, paired-test summaries, and figure provenance bundle. A neutral replay should rebuild the vocabulary selector row sets, run the Qwen3-8B benchmark matrix over the same frozen task ids, rebuild the result tables, and then rebuild the paper. The seed policy for the main negative claim uses three random bypass seeds; deterministic selectors have one frozen row set.

Family	p-value range	Effect range	Decision
TriviaQA rare entity	0.625–1	+0.20 to +0.40 pp	No significant random-control win
Natural Questions rare entity	1	-0.20 to 0.20 pp	No significant random-control win
MMLU	1	+0.00 to +0.00 pp	No significant random-control win
ARC-Challenge	1	+0.00 to +0.42 pp	No significant random-control win
HellaSwag	0.375–1	+0.00 to +1.25 pp	No significant random-control win
lm-eval macro	0.453–1	+0.14 to +0.42 pp	No significant random-control win

Table 3: Paired exact sign tests find 0/18 random-control comparisons with $p < 0.05$ for the entity-title plus norm selector.

For WikiText-2 backend context, the same-backend HF rows are 17.8694 PPL for FP16 and 18.1605 PPL for the best HF selector diagnostic. The GGUF context rows are 8.8315 PPL for FP16, 8.9666 for Q4-K-M, and 9.0466 for Q4-K-M with vocabulary forcing. These GGUF values are recorded only as deployment-context evidence because they use a different backend and likelihood path from the HF QA and lm-eval rows.